

ML Fundamentals — Assignment Set

3 practical mini-projects: EDA, visualization, and Linear Regression

General instructions: Every assignment below requires you to filter your dataset using your own name/roll number, as described in each assignment. This means no two students will work with an identical subset — submissions with identical results/plots to another student's will be treated as not personalized and marked accordingly. Use Python (Pandas, NumPy, Matplotlib, Seaborn, scikit-learn) in a Jupyter or Google Colab notebook and make doc file of your code and output.

Assignment 2: House Price Prediction — Bengaluru Housing Data

Dataset

Bengaluru House Price dataset (search 'Bengaluru House Price Data' on Kaggle).

Personalize your subset (mandatory — do this first)

Filter the dataset to locations whose name starts with the same letter as your first name, AND keep only rows with bathrooms $\geq$ (last digit of your roll number). If this leaves fewer than 150 rows, drop the bathroom condition and keep the location filter only.

Tasks

1. Perform EDA: check shape, data types, missing values, duplicates, and outliers in price and area.
2. Clean the data: handle missing values, convert 'total_sqft' ranges (e.g. '2100-2850') to a single numeric value, and create a new 'price_per_sqft' column.
3. Create at least 5 plots using Matplotlib/Seaborn: price distribution, boxplot of price by BHK, scatterplot of area vs price, correlation heatmap, and bar chart of average price by location. Label every axis and title, and write a 2-3 line insight under each plot.
4. Build a Linear Regression model predicting price from area, BHK, bathrooms, and location (one-hot encoded).
5. Split data 80/20 (train/test) and report R-squared and RMSE on the test set. Iterate — remove outliers, try a log-transform of price, add or drop features — until test R-squared reaches the pass condition below.
6. Write a half-page summary of which factors most affected price and why your final model performs as it does.

Pass condition

Test-set R-squared must be at least 0.75. Marks are reduced for models that hit this only by overfitting (e.g., huge gap between train and test R-squared) — mention both scores in your report.

Grading (out of 10)

Criteria	Marks
EDA & data cleaning	2
Visualizations (5+, labeled, with written insights)	3
Model performance (R-squared ≥ 0.75 on test set, with evidence of iteration)	3
Code clarity, comments, and summary write-up	2

Deliverables

Jupyter/Colab notebook (.ipynb) + 1-page PDF summary report.

Assignment 3: Used Car Price Prediction — CarDekho Dataset

Dataset

'Vehicle Dataset from CarDekho' on Kaggle.

Personalize your subset (mandatory — do this first)

Keep only cars of the fuel type assigned by (your roll number mod 3): 0 = Petrol, 1 = Diesel, 2 = CNG/LPG/Electric. If your assigned class has fewer than 150 rows, add Petrol cars until you reach at least 150 rows.

Tasks

1. Perform EDA: inspect data types, missing values, duplicate listings, and outliers in selling_price and km_driven.
2. Clean the data: engineer a 'car_age' column from the year, and encode categorical columns (fuel, seller_type, transmission, owner).
3. Create at least 5 plots using Matplotlib/Seaborn: price distribution, boxplot of price by transmission type, scatterplot of car_age vs price, scatterplot of km_driven vs price, and a correlation heatmap. Label everything and add a short written insight under each.
4. Build a Linear Regression model predicting selling_price from car_age, km_driven, fuel, transmission, and owner.
5. Split data 80/20 and report R-squared and RMSE on the test set. Improve the model — handle skew in price with a log-transform, remove extreme outliers, add/remove features — until test R-squared reaches the pass condition below.
6. Write a half-page summary of which factors reduced resale value the most in your subset.

Pass condition

Test-set R-squared must be at least 0.75. State both train and test R-squared in your report to show the model isn't just memorizing the training data.

Grading (out of 10)

Criteria	Marks
EDA & data cleaning	2
Visualizations (5+, labeled, with written insights)	3
Model performance (R-squared ≥ 0.75 on test set, with evidence of iteration)	3
Code clarity, comments, and summary write-up	2

Deliverables

Jupyter/Colab notebook (.ipynb) + 1-page PDF summary report.

Assignment 4: Student Exam Score Prediction — Study Habits Dataset

Dataset

'Student Habits vs Academic Performance' (or similar 'Student Study Performance') dataset on Kaggle.

Personalize your subset (mandatory — do this first)

Keep only rows where gender matches your own gender, AND study_hours $\geq$ (your roll number mod 5) + 1. If this leaves fewer than 150 rows, drop the study-hours condition and keep the gender filter only.

Tasks

1. Perform EDA: check missing values, duplicates, and outliers in exam scores and study hours.
2. Clean the data: handle missing values and encode categorical fields (e.g., part-time job status, internet access).
3. Create at least 5 plots using Matplotlib/Seaborn: score distribution, scatterplot of study_hours vs score, boxplot of score by part-time-job status, correlation heatmap, and a bar chart of average score by a sleep-hours bucket. Label everything and add a short written insight under each.
4. Build a Linear Regression model predicting exam score from study_hours plus at least 3 other features.
5. Split data 80/20 and report R-squared and RMSE on the test set. Refine the model — feature scaling, dropping weak/noisy features, handling outliers — until test R-squared reaches the pass condition below.
6. Write a half-page summary of which habits most affected exam performance in your subset.

Pass condition

Test-set R-squared must be at least 0.70.

Grading (out of 10)

Criteria	Marks
EDA & data cleaning	2
Visualizations (5+, labeled, with written insights)	3
Model performance (R-squared $\geq$ 0.70 on test set, with evidence of iteration)	3
Code clarity, comments, and summary write-up	2

Deliverables

Jupyter/Colab notebook (.ipynb) + 1-page PDF summary report.

Note for the instructor

All three assignments are captured in the 'AssignmentPractical' tab of the syllabus workbook (2026-30_Syllabus_Template.xlsx), matching the institute's standard columns. This document is the classroom-ready handout version of the same three assignments.